From microscopic to macroscopic models of tissue

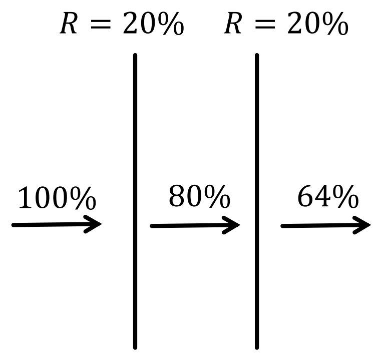
The nature of light – what do we really need to know?

- Light is a wave
- Light has polarization
- Interference

Alhazen did not know all of that, but made big contributions to optics 1,000 years ago

Light in our chaotic world – the Fabry Pérot interferometer as an example

The naïve approach:



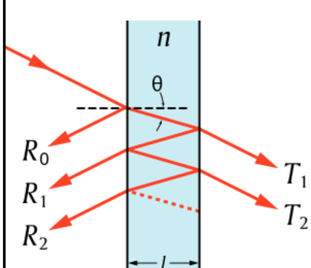
The total transmission of **power** is $(1-R)^2$

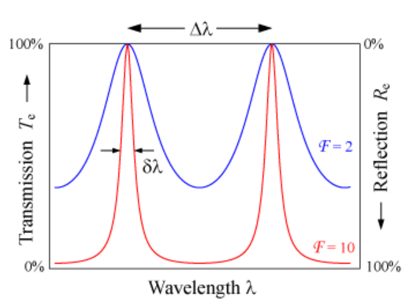
Light in our chaotic world – the Fabry Pérot interferometer as an example

A full analysis must include interference effects:

$$T = \frac{(1-R)^2}{1 + R^2 - 2R\cos\delta}$$

$$\delta = \left(\frac{2\pi}{\lambda}\right) 2n\ell\cos\theta$$





$$mean(T) = \frac{(1-R)^2}{1-R^2}$$

When is the interference averaged out?

$\delta = \left(\frac{2\pi}{\lambda}\right) 2n\ell$ Let's assume: λ=700 nm, l~25 μm, n=1.33

If ℓ changes by λ/2n ~ 270 nm

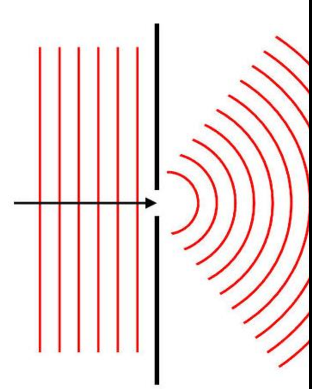
If λ changes by ~ 7 nm

λ=700 nm λ=707 nm

Averaging out of interference effects can be done temporally, spatially, or spectrally

Does light have a direction?

- Only plane waves have a unique direction
- The direction of a beam or ray still has meaning if its cross-section is much larger than the wavelength (ray optics)
- To understand the angular distribution of the beam's direction, we look in the far field



How we look at light in diffuse optics

- We sum intensities and *not* amplitudes because interference effects are averaged out
- We look at "large" volumes/areas to define directionality
- Locally, scattering effects are assumed weak
- Direction is understood in the context of the far-field
- We assume unpolarized light

Power conservation for light

EM theory provides an energy-conservation relation – Poynting's Theorem

$$\frac{\partial}{\partial t}\iiint_V W dV + \iint_{\partial V} \mathbf{S}\cdot dA + \iiint_V J\cdot E dV = 0$$

$$\frac{\partial W}{\partial t} + \nabla\cdot\mathbf{S} + \mathbf{J}\cdot\mathbf{E}dV = 0$$

$$\mathbf{S} = \frac{c}{4\pi}\mathbf{E}\times\mathbf{H}$$

$$W = \frac{1}{8\pi}(\epsilon\mathbf{E}^2 + \mu\mathbf{H}^2)$$

Power conservation for light

In the far field (or for planar waves), the magnetic and electric fields are orthogonal, leading to a simplified equation for the Poynting vector.

$$\frac{\partial W}{\partial t} + \nabla \cdot \mathbf{S} + \mathbf{J} \cdot \mathbf{E} dV = 0$$

$$\frac{1}{c}\frac{\partial |<\mathbf{S}>|}{\partial t} + \nabla \cdot <\mathbf{S}> + \alpha <\mathbf{S}> = 0$$

$$\mathbf{S} = \frac{c}{4\pi}\mathbf{E} \times \mathbf{H}$$

$$W = \frac{1}{8\pi}(\epsilon \mathbf{E}^2 + \mu \mathbf{H}^2)$$

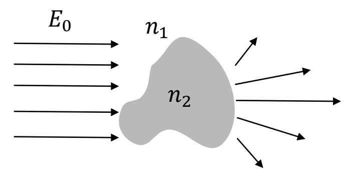
$$\mathbf{J} \propto \mathbf{E}$$

Absorption of a single particle

- Absorption efficiency: $Q_a^{eff} = \frac{P_{abs}}{P_{inc}} = \frac{P_{abs}}{<\mathbf{S}_{inc}>}\frac{1}{A} = \frac{\sigma_a}{A}$, where $\sigma_a = \frac{P_{abs}}{<\mathbf{S}_{inc}>}$ is the absorption cross-section
- The absorption cross-section is a fraction of the geometrical cross-section, e.g. a particle with a cross-section A that absorbs 50% of the energy will have an absorption cross-section of 0.5A

Scattering of a single particle

$$E(r) = E^{(inc)}(r) + E^{(sc)}(r)$$



$$E^{(inc)}(r) = E_0 \exp(ik\hat{\mathbf{s}}_0 \cdot \mathbf{r})$$

$$E^{(sc)}(r) = E_0 \mathrm{f}(\hat{\mathbf{s}}, \hat{\mathbf{s}}_0) \frac{\exp(ikr)}{r}$$

$\mathrm{f}(\hat{\mathbf{s}}, \hat{\mathbf{s}}_0)$ - the scattering amplitude

$$< \mathbf{S} >^{(sc)} = \mathbf{S}^{(inc)} \frac{|\mathrm{f}(\hat{\mathbf{s}}, \hat{\mathbf{s}}_0)|^2}{r^2} \hat{\mathbf{s}}$$

Scattering and total cross-sections

- The scattering cross-section quantifies all the energy that changed its direction:

$$\sigma_{sc} = \iint_S \frac{|\mathrm{f}(\hat{\mathbf{s}}, \hat{\mathbf{s}}_0)|^2}{r^2} \hat{\mathbf{s}} = \iint_{4\pi} |\mathrm{f}(\hat{\mathbf{s}}, \hat{\mathbf{s}}_0)|^2 d\Omega$$

- The total cross-section: $\sigma_{tot} = \sigma_a + \sigma_{sc}$

"Forward scattering theorem" or "optical theorem"

- In the far field:

$$E(r) \cong E_0 \exp(ik\hat{\mathbf{s}}_0 \cdot \mathbf{r}) + E_0 \mathrm{f}(\hat{\mathbf{s}},\hat{\mathbf{s}}_0)\frac{\exp(ikr)}{r}$$

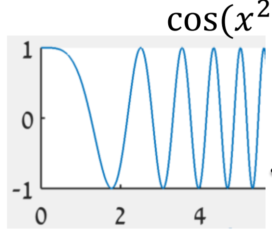
- Optical theorem:

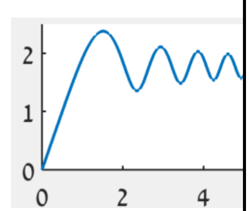
$$\frac{4\pi}{k}\mathrm{Im}[\mathrm{f}(\hat{\mathbf{s}}_0,\hat{\mathbf{s}}_0)] = \sigma_{tot}$$

Illustration for the stationary phase approximation

$$f(a) = \int_{-a}^{a} e^{-ix^2}dx$$

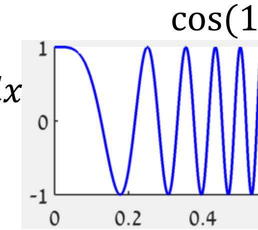
$\cos(x^2)$

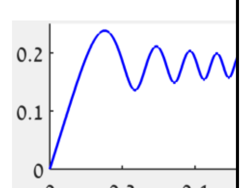




$$g(a) = \int_{-a}^{a} e^{-100ix^2}dx$$

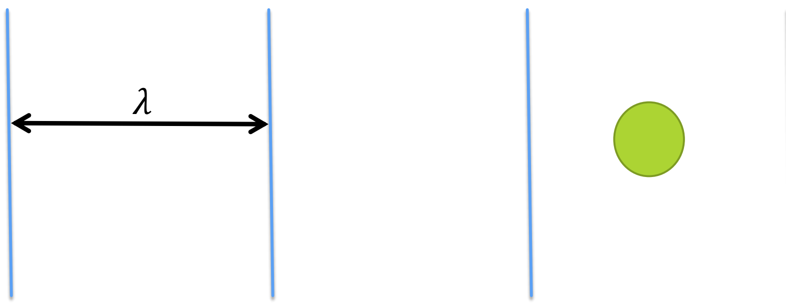
$\cos(100x^2)$





Rayleigh scattering

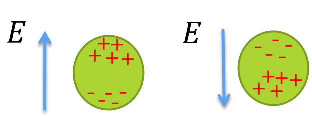
Assumptions: particle much smaller than wavelength

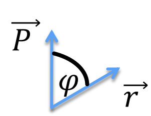


We can assume that the electric field inside the particle is quasi-static -> similar to a particle in an alternating electric field = dipole

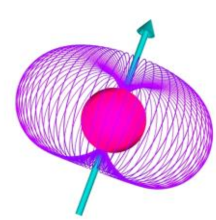
Rayleigh scattering

Polarization 1:





$$E_{dip} = P\, k^2 \sin\varphi \frac{\exp(-ikr)}{r}$$



Polarization 2:

E ⊗ E ⊙

$$\varphi = 90$$

$$E_{dip} = P\, k^2 \frac{\exp(-ikr)}{r}$$

$$< \mathbf{S} >^{(sc)} \propto \frac{1+(\cos\theta)^2}{\lambda^4}\frac{1}{r^2} \Rightarrow < \mathbf{S} >^{(sc)} \propto \frac{1+(\cos\theta)^2}{\lambda^4}\frac{1}{r^2}V^2$$

Mie theory

- Exact solution to the scattering of EM plane wave from a sphere
- The solution is given by an infinite series of spherical harmonics
- Most important insight: the larger the scattering particle is, the more forward scattering it is

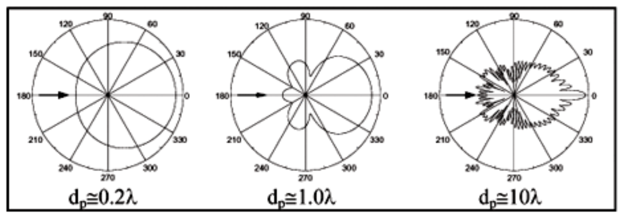
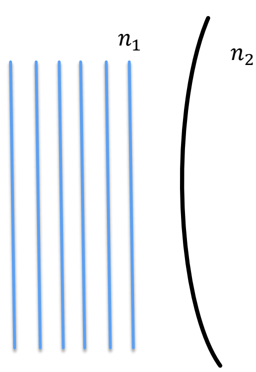


Figure 17. Mie scatter diagram for different size seeding particles dp is particle size.

K. D. Jensen, J. Braz. Soc. Mech. Sci.& Eng. **26**, 2004.

Large particles



- Scattering cross-section in reflection is small: $\propto \frac{(n_2 - n_1)^2}{(n_2 + n_1)^2}$
- In transmission, refraction deflects the beam => most of the intensity is found at small angles.
- For infinitely large particles: $\sigma_{tot} \rightarrow 2A$, the extinction paradox

The phase function

$$p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0) = \frac{|\mathrm{f}(\hat{\mathbf{s}},\hat{\mathbf{s}}_0)|^2}{\sigma_{tot}}$$

- Represents the intensity of light scattered in a certain direction as a fraction of the total loss.
- Imported nomenclature ("phase" refers to lunar phases) – has nothing to do with the phase of the EM wave
- The albedo or whiteness: $W_0 = \iint_{4\pi} p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0)\mathrm{d}\Omega = \frac{\sigma_s}{\sigma_{tot}}$

Common phase functions

- Isotropic:

$$p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0) = \frac{W_0}{4\pi}$$

- Rayleigh:

$$p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0) = p(\hat{\mathbf{s}}\cdot\hat{\mathbf{s}}_0) = \frac{3}{16\pi}(1 + (\hat{\mathbf{s}}\cdot\hat{\mathbf{s}}_0)^2)W_0$$

- Henyey-Greenstein:

$$p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0) = p(\hat{\mathbf{s}}\cdot\hat{\mathbf{s}}_0) = \frac{1}{4\pi}\frac{W_0(1-g^2)}{(1+g^2-2g\hat{\mathbf{s}}\cdot\hat{\mathbf{s}}_0)^{3/2}}$$

Henyey-Greenstein's phase function

- Originally constructed for interstellar scattering
- Found to <u>*statistically*</u> represent scattering in tissue (not an analytical solution to a specific scatterer)
- Enable anisotropy

L.G. Henyey, J.L. Greenstein, Diffuse radiation in the galaxy, Astrophysical Journal .1941 ,93:70-83

Henyey-Greenstein's phase function

$$p(\hat{\mathbf{s}}, \hat{\mathbf{s}}_0) = p(\hat{\mathbf{s}} \cdot \hat{\mathbf{s}}_0) = \frac{1}{4\pi} \frac{W_0(1-g^2)}{(1+g^2-2g\hat{\mathbf{s}} \cdot \hat{\mathbf{s}}_0)^{3/2}}$$

Where *g* is the anisotropy factor:

$$g = \frac{1}{W_0} \int_{(4\pi)} p(\hat{\mathbf{s}}, \hat{\mathbf{s}}_0) \hat{\mathbf{s}} \cdot \hat{\mathbf{s}}_0 \mathrm{d}\Omega$$

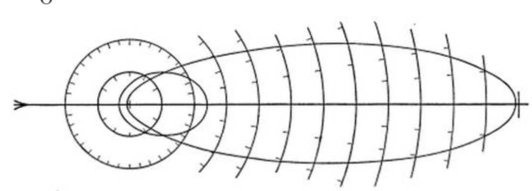
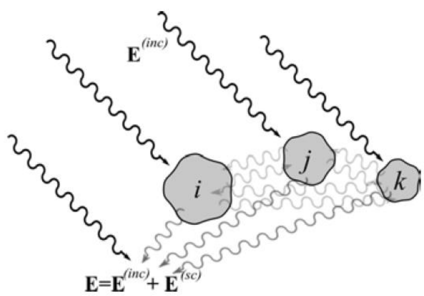


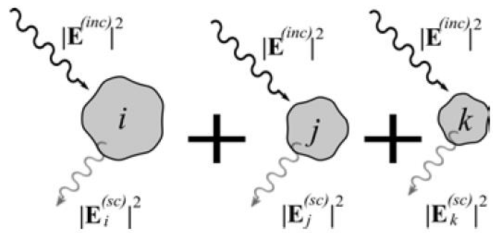
FIG. 3.—Polar diagram of the phase function of equation (2) The more elongated curve is for $g = +\frac{2}{3}$; the other, for $g = +\frac{1}{3}$. The radiation is incident on the particle from the left, as shown by the arrow.

Multiple scatterers and absorbers

Full picture:



Our approximation:

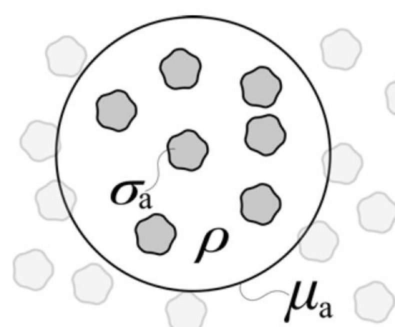


$$\bar{\sigma}_{sc} = \sum_{i=1}^{N} \sigma_{sc}^{(i)} \qquad \bar{\sigma}_a = \sum_{i=1}^{N} \sigma_a^{(i)}$$

Scattering and absorption coefficients

Scattering coefficient:

$$\mu_s = \rho\sigma_{sc} \quad \left(cm^{-1}\right)$$

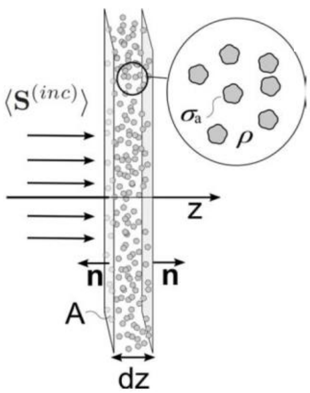


Absorption coefficient:

$$\mu_a = \rho\sigma_a \quad \left(cm^{-1}\right)$$

Where ρ is the density of particles: $\rho = N/V$

Beer Lambert law – absorption in non-scattering slab



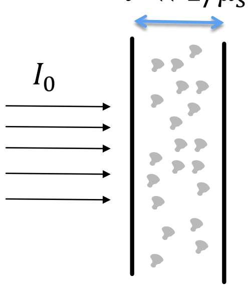
$$[S(z=0) - S(z=dz)]\, A = -N\sigma_a \|\mathbf{S}^{(inc)}\|$$

Substituting: $N = \rho V = \rho A dz,$

We obtain $\dfrac{dS}{dz} = -\underbrace{\rho\sigma_a}_{\mu_a} S,$

$$S(z=L) = S^{(inc)} \exp\left(-\mu_a L\right)$$

The effect of μ_s



Detector in far field

$$I_d = I_0 - \mu_s \ell I_0 \quad \Rightarrow \quad \frac{\Delta I}{\Delta z} = -\mu_s I$$

A differential equation cannot be simply constructed

The reduced scattering coefficient

Reduced scattering coefficient:

$$\mu_s' = \mu_s(1-g)\left(cm^{-1}\right)$$

- Represents the rate in which light losses its directionality
- Derived from the full propagation equations
- As expected, increase in anisotropy (towards 1) leads to reduction in the reduced scattering coefficient

Scattering (reduced) mean free path

$l_s = (\mu_s)^{-1} \approx 0.1$ mm

l_s'

The scattering strength is expressed in terms of the scattering mean free path l, which is the average distance waves propagate between scattering events.

We may characterize a homogeneous scattering medium by its scattering mean free path l, which specifies the average distance travelled by a photon between two scattering events.

The scattering mean free path is also traditionally understood as the *average* distance light needs to travel until it suffers the next scattering event. Great care should be taken when using this definition.

A. P. Mosk Nature Photonics **6** (2012)
R. Horstmeyer et al. Nature Photonics **9** (2015)

J. Ripoll "Principles od diffuse light propagation" (World Scientific Publishing 2012)

Reduced scattering coefficient of tissue

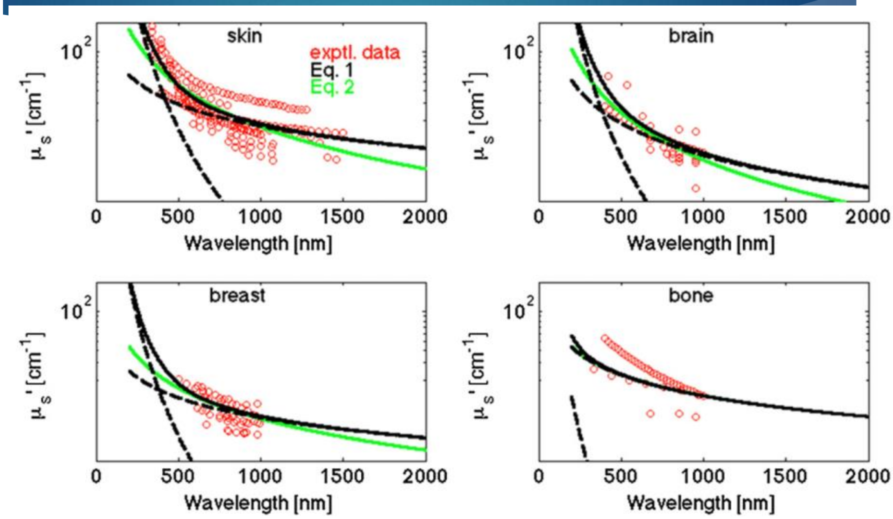
Curved fitting approach:

$$\mu_s' = a\left(\frac{\lambda}{500\ (\mathrm{nm})}\right)^{-b} \qquad (1)$$

$$\mu_s'(\lambda) = a'\left(f_{\mathrm{Ray}}\left(\frac{\lambda}{500\ (\mathrm{nm})}\right)^{-4} + (1 - f_{\mathrm{Ray}})\left(\frac{\lambda}{500\ (\mathrm{nm})}\right)^{-b_{\mathrm{Mie}}}\right). \qquad (2)$$

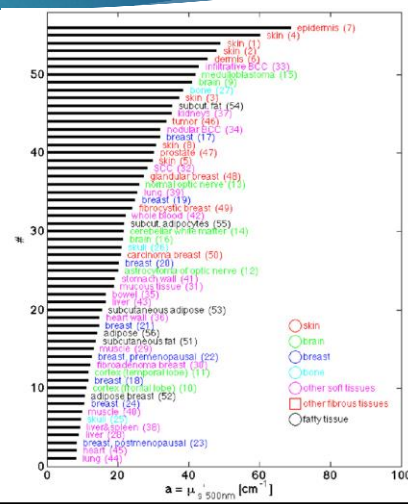
S. L. Jacques, Phys. Med. Biol. 58, R37–R61 (2013)

Reduced scattering coefficient of tissue



Tissue heterogeneity

- Values change among organs
- Values vary among studies
- "Typical" average values: $\mu'_s = 10 - 20$ cm^{-1} in the NIR



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